INCH-POUND
MIL-DTL-24643/84
27 November 2023

DETAIL SPECIFICATION SHEET

CABLE, ELECTRICAL, -4 DEGREES FAHRENHEIT (-20 DEGREES CELSIUS) TO +176 DEGREES FAHRENHEIT (+80 DEGREES CELSIUS), 150 VOLTS, TOPSIDE, TYPE LSEC6FS (INCLUDING VARIANTS LSEC6OS AND LSEC6OSW)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24643.

This cable can be manufactured in three variations: LSEC6FS (Metallic Foil Shield Non-Watertight), LSEC6OS (Metallic Foil and Braid Shielded Non-Watertight), and LSEC6OSW (Metallic Foil and Braid Shielded Watertight).

Construction (Watertight and Non-Watertight)

First - Copper conductor, solid, uncoated, in accordance with ASTM B3, 23 AWG, 0.0226 inch nominal. Conductor diameter may vary from nominal as necessary in order for the completed cable to meet the specified electrical requirements.

Second - Solid or foamed polyolefin insulation. Wall thickness as necessary, in order for the completed cable to comply with the electrical requirements specified herein. The maximum insulation diameter shall not exceed 0.051 inch.

Third - Two insulated conductors, twisted together with an appropriate lay length for each pair such that the completed cable complies with the electrical requirements specified herein. Wire identification codes shall be in accordance with ANSI/TIA 568.2 as follows:

Pair 1 - White/blue and blue

Pair 2 - White/orange and orange

Pair 3 - White/green and green

Pair 4 - White/brown and brown

Optional identification codes for cables with tightly twisted pairs less than 1.5 inches (38 millimeters) are as follows:

Pair 1 - White and blue

Pair 2 - White and orange

Pair 3 - White and green

Pair 4 - White and brown

Fourth - Four pairs shall be cabled together with a lay as required to comply with the electrical requirements specified herein. Cabling sequence shall be consecutive. Fillers or other means may be used as necessary to form a firm, well-rounded cable that complies with the electrical requirements specified herein. See table I for overall diameter.

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Fifth - An optional binder.

Construction for Variant Type LSEC6FS

Sixth - Copper drain wire, tin-coated, 24 AWG, solid or stranded, in accordance with ASTM B286.

Seventh - Metallic foil/polyester tape (metallic foil side facing in and in contact with the drain wire). Tape may be longitudinally or helically applied, with a minimum overlap of 10 percent.

Eighth - An optional binder.

Ninth - Cross-linked polyolefin jacket. Absolute minimum average wall thickness 0.040 inch. See table I for cable overall diameter.

Tenth - Copper foil/polyester tape. Applied helically with 50 percent minimum overlap (metallic foil side facing out and in contact with the braid).

Eleventh - Braided shield of tin-coated copper. Braid shall be constructed such that the completed cable complies with the surface transfer impedance requirements specified herein. In addition, the minimum coverage of the braid shall be 90 percent and the minimum shield strand size shall be 34 AWG.

Twelfth - Copper foil/polyester tape/copper foil. Applied helically with 50 percent minimum overlap. Tape shall provide electrical contact between the inner and outer braided shields.

Thirteenth - Braided shield of tin-coated copper. Braid shall be constructed such that the completed cable complies with the surface transfer impedance requirements specified herein. In addition, the minimum coverage of the braid shall be 90 percent and the minimum shield strand size shall be 34 AWG.

For watertight constructions only, the use of solid or semi-solid water-block materials (such as silicone, hot melts, or gels) is not allowed on the outer braid. Watertightness shall be by means that could include, but are not limited to, water-swell yarns incorporated within or over the braid or water-swell tape helically applied over the outer braid with 10 percent minimum overlap. The water-block material shall not stick or adhere to the outer braid strands.

Fourteenth - An optional binder (tape shall not be clear).

Fifteenth - Cross-linked polyolefin jacket. Absolute minimum average wall thickness 0.050 inch. See table I for cable overall diameter.

NOTE: Manufacturer's identification tape is required (see 3.4.9 of MIL-DTL-24643).

Construction for Variant Types LSEC6OS and LSEC6OSW

Sixth - Metallic foil/polyester tape (metallic foil side facing out and in contact with the braid). Tape may be longitudinally or helically applied, with a minimum overlap of 10 percent.

Seventh - Braided shield of 36 AWG or 34 AWG tin-coated copper.

Eighth - An optional binder.

Ninth - Cross-linked polyolefin jacket. Absolute minimum average wall thickness 0.040 inch. See table I for cable overall diameter.

Tenth - Copper foil/polyester tape. Applied helically with 50 percent minimum overlap (metallic foil side facing out and in contact with the braid).

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Eleventh - Braided shield of tin-coated copper. Braid shall be constructed such that the completed cable complies with the surface transfer impedance requirements specified herein. In addition, the minimum coverage of the braid shall be 90 percent and the minimum shield strand size shall be 34 AWG.

Twelfth - Copper foil/polyester tape/copper foil. Applied helically with 50 percent minimum overlap. Tape shall provide electrical contact between the inner and outer braided shields.

Thirteenth - Braided shield of tin-coated copper. Braid shall be constructed such that the completed cable complies with the surface transfer impedance requirements specified herein. In addition, the minimum coverage of the braid shall be 90 percent and the minimum shield strand size shall be 34 AWG.

Fourteenth - An optional binder (tape shall not be clear).

Fifteenth - Cross linked polyolefin jacket. Absolute minimum average wall thickness 0.050 inch. See table I for cable overall diameter.

NOTE: Manufacturer's identification tape is required (see 3.4.9 of MIL-DTL-24643).

TABLE I. Details.

Military part no. M24643/84	Type and size	Overall diameter (inch)	
		min.	max.
-01UD	LSEC6FS-4	0.530	0.595
-02UD	LSEC6OS-4	0.550	0.605
-03UD	LSEC6OSW-4	0.580	0.645

TABLE II. Swept electrical parameters.

Frequency (MHz)	Return loss dB/328 feet (min.)	Insertion loss dB/328 feet (max.)	NEXT dB/328 feet (min.)	PSNEXT dB/328 feet (min.)	ACRF dB/328 feet (min.)	PSACRF dB/328 feet (min.)
1.0	20.0	2.1	74.3	72.3	67.8	64.8
4.0	23.0	3.8	65.3	63.3	55.8	52.8
8.0	24.5	5.3	60.8	58.8	49.7	46.7
10.0	25.0	5.9	59.3	57.3	47.8	44.8
16.0	25.0	7.5	56.2	54.2	43.7	40.7
20.0	25.0	8.4	54.8	52.8	41.8	38.8
25.0	24.3	9.4	53.3	51.3	39.8	36.8
31.25	23.6	10.5	51.9	49.9	37.9	34.9
62.5	21.5	15.0	47.4	45.4	31.9	28.9
100.0	20.1	19.1	44.3	42.3	27.8	24.8
200.0	18.0	27.6	39.8	37.8	21.8	18.8
250.0	17.3	31.1	38.3	36.3	19.8	16.8
300.0	16.8	34.3	37.1	35.1	18.3	15.3
400.0	15.9	40.1	35.3	33.3	15.8	12.8
500.0	15.2	45.3	33.8	31.8	13.8	10.8

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TABLE III. Additional swept electrical parameters.

Frequency (MHz)	TCL dB (min.)	ELTCTL dB/328 feet (min.)	Propagation delay nS/328 feet (nominal)	Delay skew nS/328 feet (max.)
1.0	40.0	35.0	570	45
4.0	40.0	23.0	552	45
8.0	40.0	16.9	547	45
10.0	40.0	15.0	545	45
16.0	38.0	10.9	543	45
20.0	37.0	9.0	542	45
25.0	36.0	7.0	541	45
30.0	n/S	5.5	n/S	n/S
31.25	35.1	n/S	540	45
62.5	32.0	n/S	539	45
100.0	30.0	n/S	538	45
200.0	27.0	n/S	537	45
250.0	26.0	n/S	536	45
300.0	25.2	n/S	536	45
400.0	24	n/S	536	45
500.0	23	n/S	536	45
NOTE: Values are reference points. Compliance shall be determined utilizing the formulas in ANSI/TIA-568.2.				

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REQUIREMENTS:

Qualification required.

INSPECTION:

Basic electricals:

Conductor resistance (ohms/328 feet at 68 °F [20 °C], max.)	9.38
Conductor resistance unbalance (percent, max.)	4
Voltage withstand	
Conductor to conductor	1.5 kVAC or 2.5 kVDC
Conductor to overall shield	1.5 kVAC or 2.5 kVDC
Inner overall shield to outer overall shield	3.5 kVAC or 4.5 kVDC
Insulation resistance (megohms/328 feet, max.)	Not applicable
Jacket flaws	No failure
Conductor and shield continuity	No failure

Group A:

Visual and dimensional	No failure
Capacitance	
Mutual capacitance (nominal at 1 kilohertz [kHz]) (nanofarad [nF]/328 feet)	5.6
Pair-to-ground capacitance unbalance (max. at 1 kHz, picofarad [pF]/328 feet)	330
Characteristic impedance	No requirement
Watertightness (type LSEC6OSW only) (see MIL-DTL-24643 for limits of water leakage)	No failure
Swept electrical parameters	See table II and table III

Group B:

Cold bending (cable)	No failure
Drip 203±1.8 °F (95± 1 °C) (type LSEC6OSW only)	No failure
Physicals (unaged)	
Insulation (extruded insulation only)	
Tensile strength (lb/in^2, min.)	450
Elongation (percent, min.)	75

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Inner and outer jacket (cable)	
Tensile strength (lb/in^2, min.)	1,300
Elongation (percent, min.)	160
Tear (lb/in thickness, min.)	35
Thermoset proof test (percent, max.)	
Insulation	Not applicable
Inner and outer jacket (when tested at 392 °F [200 °C])	50

Group C:

Physicals (aged) (air oven)	
Insulation (extruded insulation only)	
Tensile strength (percent of unaged, min.)	75
Elongation (percent of unaged, min.)	75
Inner and outer jacket (cable)	
Tensile strength (percent of unaged, min.)	60
Elongation (percent of unaged, min.)	60
Shrinkage	No failure
Cable fill (sealant) removability (applicable to watertight constructions only)	No failure
Heat distortion (percent of unaged, max.)	30
Permanence of printing (jacket) (cycles, min.)	125
Shield (conformance to material, construction, and coverage)	No failure
Surface transfer impedance (outer shields only)	
Milliohms per meter (max.)	5
Electromagnetic pulse (EMP) response (dB, min.)	80

Group D:

Flame propagation (cable)	No failure

Qualification inspection:

Qualification inspection shall include basic electricals; groups A, B, C, and D; plus the following:

Aging and compatibility (cable) (203±3.6 °F [95±2 °C])	No failure
Abrasion resistance (jacket) (scrapes, min.)	75
Acid gas equivalent (percent, max.)	

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Inner and outer jacket	2
Fillers	2
Insulation	18
Halogen content (percent, max.)	
Inner and outer jacket	0.2
Fillers	0.2
Insulation	0.2
Immersion (inner and outer jacket)	
Tensile strength (percent of unaged, min.)	50
Elongation (percent of unaged, min.)	50
Smoke index (max.)	
Inner and outer jacket	25
Fillers	45
Insulation	45
Toxicity index (max.)	
Inner and outer jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) – type A (hardness, min.)	80
Weathering (inner and outer jacket)	No failure
Stress resistance (types LSEC6OSW and LSEC6OS)	No failure
Temperature humidity (type LSEC6OSW only)	No failure
Electrical moisture absorption	Not applicable
Long-term inner and outer jacket immersion	No failure

UNIT ORDERING LENGTHS:

Type and size	Feet (nominal)
All	500

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CONCLUDING MATERIAL

Custodians:
Army – MI
Navy – SH

Preparing activity:
Navy – SH
(Project 6145-2017-005)

Review activities:
Army – AR, CR
Navy – CG, EC
DLA – CC

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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